CIRR-MoE: Cross-Inductive Reliability-Guided Mixture of Experts for Decision-Specific Trustworthy Industrial Prediction

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ABSTRACT

Industrial predictive-maintenance models are commonly evaluated by aggregate accuracy or regression error, yet these measures do not answer three distinct deployment questions: which predictive expert is reliable for the current operating condition, whether the final prediction itself is likely to fail, and whether the observation is outside the familiar operating regime. We introduce CIRR-MoE, a heterogeneous mixture-of-experts framework that treats these questions as separate reliability problems rather than compressing them into one generic confidence score. The method first obtains genuine out-of-fold predictions from XGBoost, CatBoost, LightGBM, ExtraTrees and histogram gradient boosting. Their held-out errors define a cross-inductive reliability representation from which sample-specific expert competence $R_{\mathrm{C}}(x,m)$ is estimated. Experts are then routed through temperature-controlled reliability weights and fused for regression or classification. Downstream trust is decomposed into final-prediction failure risk R_{F}, open-world unfamiliarity R_{O}, and reliability-conditioned conformal uncertainty, enabling selective prediction, abstention and unknown-condition rejection. On the 55,999-sample EngineFaultDB benchmark, CIRR-MoE achieved $R^2 = 0.99696$, RMSE $= 51.42646$ and MAE $= 38.27975$. On external C-MAPSS FD001 remaining-useful-life prediction it achieved $R^2 = 0.77130$ and RMSE $= 19.87277$, while PHM 2009 gearbox classification reached macro-F1 $= 0.93651$. Failure-risk AUROC reached 0.79905, classification error-awareness AUROC 0.94074, and true-unknown AUROC reached 0.90822 on Tennessee Eastman and 0.82448 on PHM. Reliability-conditioned conformal prediction achieved 0.90964 empirical coverage while improving the highest-risk quartile coverage to 0.89600. Comprehensive component and expert-removal ablations further show that cross-fitting, reliability routing, explicit failure modeling and open-world modeling make distinct contributions. These results support a decision-specific view of trustworthy industrial prediction in which expert competence, prediction failure and unfamiliarity are modeled as related but non-interchangeable quantities.

Introduction

Unexpected equipment degradation remains a persistent source of production loss, maintenance cost and safety risk across manufacturing, transportation, energy and process industries. Predictive maintenance seeks to move intervention from reactive repair toward condition-informed scheduling by learning actionable relationships between sensor observations and future or current equipment health[1,2,4]. The rapid growth of industrial sensing and data acquisition has consequently shifted fault diagnosis and prognostics from manually engineered signal indicators toward machine-learning systems that can learn nonlinear associations from heterogeneous measurements[3,5]. This evolution has produced highly accurate models, but it has also changed the nature of the reliability problem. When a model is deployed in an industrial workflow, high average accuracy is insufficient if it remains unable to recognize the samples for which its own prediction is unreliable.

This issue is particularly important for structured industrial data. Gradient-boosted trees and related ensembles remain formidable baselines for tabular prediction[22,23,29,38,39,54]. Their strong performance has encouraged hybrid ensembles, stacking and mixture-of-experts (MoE) designs intended to exploit model complementarity[14,25–27]. However, most ensembles answer a global optimization question: which combination of models minimizes average validation loss? Fixed averaging and stacking provide one set of global combination weights, whereas conventional learned MoE gates are trained primarily to improve task loss. Neither formulation necessarily establishes whether the gate value for expert m can be interpreted as a calibrated estimate of that expert's local competence on a specific sample.

At the same time, the trustworthy-AI literature in fault diagnosis is expanding rapidly. Recent studies quantify epistemic

uncertainty, calibrate Bayesian diagnostic models, adaptively fuse multi-source sensors, or construct robust interval-valued monitoring statistics[8–11]. Conformal prediction has also been introduced into process diagnosis to obtain finite-sample coverage and transparent rejection behavior[12,13]. Open-set fault diagnosis has become a distinct research direction because closed-set classifiers can assign high confidence to fault modes that were never represented during training[18–21]. These developments collectively show that reliable industrial AI must account for uncertainty and unfamiliarity. Yet they also reveal a fragmentation problem: expert selection, final-prediction failure, calibration and open-set rejection are often treated as separate techniques, or alternatively are all forced through a single uncertainty score.

We argue that these are not equivalent reliability questions. First, *expert competence* asks which model should receive predictive responsibility for the current sample. Second, *final-prediction failure risk* asks whether the fused decision is likely to be wrong even after routing. Third, *open-world unfamiliarity* asks whether the sample lies beyond the known operating/fault support. The distinction is operationally important. An expert may be relatively best among available experts but the entire expert bank may still be unreliable; a classifier may be uncertain on a difficult known sample without the sample being out-of-distribution; and an OOD detector may identify a new regime even when the predictive entropy is deceptively low. Recent reviews of trustworthy process monitoring and open-set diagnosis explicitly identify distribution inconsistency, scarce fault support, unknown fault classes and uncertainty quantification as open challenges[6,7,21].

Research gap and study hypothesis

The central gap addressed in this study is therefore not simply the absence of another ensemble model. It is the absence of a unified industrial prediction framework that (i) learns *sample-specific expert competence* from genuine held-out errors, (ii) uses that competence to route a heterogeneous expert bank, and (iii) preserves separate downstream reliability quantities for final failure and unfamiliarity. Although adaptive MoE has recently been explored for joint RUL prediction and fault diagnosis[14], and MoE has been coupled with conformal inference for reliable soft sensing[15], these approaches do not directly formulate cross-fitted expert-error prediction as the gating target or decompose trust into competence, final failure and open-world unfamiliarity. Likewise, Bayesian or evidential fault-diagnosis frameworks quantify uncertainty[8,18], but they are not designed to decide which of several heterogeneous tabular predictors should be trusted on a sample-by-sample basis.

We therefore test the hypothesis that an explicit cross-inductive reliability representation can make heterogeneous expert routing more informative than prediction-only gating or static fusion, while the resulting trust representation can support failure-aware abstention, adaptive conformal uncertainty and open-world rejection without conflating their objectives.

Contributions

The main contributions are summarized as follows.

- **Cross-inductive reliability-guided heterogeneous MoE.** We construct expert-specific reliability targets from out-of-fold (OOF) prediction errors and use the predicted errors, rather than a generic task-loss gate, to derive sample-dependent mixture weights.
- **Direct expert-competence evaluation.** Beyond final predictive metrics, the router is evaluated using exact expert-selection accuracy, top-2 competence recall, error-ranking correlation and routing regret, providing direct evidence about whether the reliability layer identifies locally suitable experts.
- **Decision-specific reliability decomposition.** We explicitly distinguish expert competence R_{C}, final-prediction failure risk R_{F} and open-world unfamiliarity R_{O}. Each quantity is linked to a different operational action: routing, accept/defer and unknown-condition rejection.
- **Trust-aware uncertainty and selective prediction.** The framework uses reliability-conditioned conformal inference and risk-based selective prediction so that uncertainty influences the deployment decision rather than being reported only as a post-hoc score.
- **Cross-task industrial validation.** The study combines internal engine-state regression, multiclass engine-fault classification, external C-MAPSS RUL regression, PHM gearbox classification, Tennessee Eastman open-world fault recognition, sensor-corruption stress tests and comprehensive component ablations.

Related Work

Data-driven fault diagnosis and prognostics

The development of intelligent fault diagnosis has progressed from manually designed time/frequency features and shallow classifiers toward learned representations, transfer learning and task-specific deep architectures. Lei et al.[3] provide a widely used roadmap for this progression, while predictive-maintenance reviews emphasize the need to connect condition monitoring with

maintenance decisions rather than merely improving benchmark accuracy[1,2,4]. Remaining-useful-life estimation has followed a similar path, with deep sequence and convolutional models increasingly replacing handcrafted degradation indicators[5,36]; recent work has also examined RUL prediction under partial sensor failure[17]. C-MAPSS remains one of the canonical RUL benchmarks because the run-to-failure simulations expose multivariate sensor trajectories under controlled degradation[34,37].

A recurring limitation of this literature is that prediction quality is often summarized by one global error measure. Such reporting can hide highly non-uniform error across operating regions. This is precisely the setting in which selective prediction and local reliability estimation become useful: an industrial system should not only minimize mean error, but should identify when its own error is likely to be large.

Tabular learning, heterogeneous ensembles and mixture-of-experts

Tree ensembles remain strong on structured data because they naturally capture nonlinear thresholds, interactions and heterogeneous feature scales[22–24,38,39]. Modern tabular deep learning has narrowed the gap through attention, parameter-efficient ensembling and foundation-model pretraining[28,29,53,54], but no single model family dominates all industrial tasks. This motivates heterogeneous ensembles in which inductive biases are deliberately diversified.

Stacked generalization learns a second-level predictor from base-model outputs[25]; classical mixtures of experts instead use a gating network to allocate samples among local experts[26,27]. Sparse MoE later became a scaling mechanism for neural networks[40,41]. In industrial prognostics, Zhou and Wang[14] recently proposed adaptive MoE for multi-task RUL prediction and diagnosis, while Meng et al.[15] coupled MoE with conformal inference for reliable soft sensing. These studies establish the relevance of adaptive expert allocation, but the gate is still generally optimized as part of predictive learning. CIRR-MoE instead makes expert error itself the supervised routing target: it learns a vector of expected held-out expert losses and converts those losses into mixture weights.

Uncertainty, calibration and trustworthy fault diagnosis

A predictive probability is not automatically a calibrated probability. Calibration research has shown that even high-performing neural models may systematically overstate confidence[42]. Bayesian approximations, deep ensembles and uncertainty decomposition provide alternative routes to epistemic and aleatoric uncertainty[30,43–45]. Their behavior can degrade under distribution shift[46], motivating explicit stress testing rather than clean-data calibration alone.

This concern is now central in fault diagnosis. Li et al.[8] combine Bayesian deep learning with confidence calibration for machinery diagnosis and OOD detection, while Bayesian uncertainty has also been used to guide transfer learning for rotating-component diagnosis[16]. Yu et al.[9] use uncertainty to weight multi-source sensor evidence, while Liu et al.[11] model epistemic uncertainty in process monitoring. The 2026 review by Liu et al.[7] synthesizes this line of work and identifies small fault samples and distribution inconsistency as core sources of epistemic uncertainty. Our framework is complementary: rather than replacing expert models by one Bayesian network, it estimates the competence of several heterogeneous predictors and then builds decision-specific reliability on top of the routed mixture.

Conformal and selective prediction

Conformal prediction provides distribution-free finite-sample coverage under exchangeability and has become a practical tool for reliable machine learning[31–33,50]. For fault diagnosis, Heddoub et al.[13] demonstrate uncertainty-aware conformal classification on industrial processes, and their later class-conditional method supports open-set rejection through empty conformal sets[12]. Selective prediction asks a different but related question: how does risk change if the model abstains on uncertain cases? Reject-option networks and risk-coverage analysis formalize this behavior[51,52]. CIRR-MoE uses failure risk as an ordering variable for selective prediction and as a scale for adaptive conformal regression intervals.

Open-world fault diagnosis

OOD detection methods based on confidence, input perturbation or energy have been extensively studied in generic machine learning[47–49]. Industrial open-set diagnosis is more specialized because unknown faults may be highly similar to known faults and because operating-condition changes can alter the feature distribution. Recent methods use evidential mining[18], dual-stage recognition with Weibull/meta-recognition mechanisms[19], and open-set domain generalization with pseudo-unknown synthesis[20]. The emerging review literature describes a transition from closed-set recognition toward open-set recognition, domain adaptation and incremental diagnosis[21]. Our objective is narrower but operationally complementary: we train a pseudo-open R_O head without exposing the true unknown classes used for final evaluation, then compare it with intrinsic uncertainty and disagreement signals. Representative recent approaches and their distinction from the proposed framework are summarized in Table 1.

Table 1. Comparison of representative recent studies in trustworthy industrial fault diagnosis, uncertainty quantification, open-set recognition and mixture-of-experts learning. The table highlights the principal reliability mechanism considered by each study and the main methodological distinction from CIRR-MoE, whose defining feature is cross-fitted, sample-specific expert-competence estimation combined with separate failure-risk and open-world reliability modeling.

Study	Year	Focus	Reliability mechanism	Distinction from CIRR-MoE
Li et al.[8]	2025	Machinery diagnosis	Bayesian UQ + calibration	Single backbone; no expert-level routing
Zhou and Wang[14]	2025	RUL + diagnosis	Adaptive MoE	Prediction-oriented gate, not OOF error routing
Heddoub et al.[13]	2025	Process diagnosis	Conformal prediction	Prediction sets without heterogeneous routing
Zhang et al.[10]	2025	Uncertain process data	Interval distributions	Measurement uncertainty rather than expert competence
Yu et al.[9]	2026	Multi-sensor diagnosis	Evidential fusion	Sensor-level rather than expert-level reliability
Heddoub et al.[12]	2026	Open-set diagnosis	Mondrian conformal	Open-set rejection without competence routing
Kong et al.[18]	2026	Open-set diagnosis	Evidence mining	Dedicated open-set model; no RC/RF decomposition
Lian et al.[19]	2026	Known/unknown faults	Dual-stage recognition	Open-set network rather than unified routing framework
Meng et al.[15]	2026	Reliable soft sensing	MoE + conformal	No OOF expert-error competence target
Liu et al.[7]	2026	Process monitoring review	Epistemic-UQ taxonomy	Review framework; no heterogeneous routing model
CIRR-MoE	–	Multi-task industrial prediction	R_C, R_F, R_O + conformal	Cross-fitted competence routing with decision-specific trust

Materials and Methods

Study design and locked-evaluation principle

The study was designed around a strict separation between predictive-model development, reliability-model development and final evaluation. The central principle is that the reliability model must learn from errors that could genuinely occur on unseen samples. Consequently, base-expert competence is never trained from the same predictions used to fit an expert. Instead, K-fold out-of-fold predictions are generated on the development data; expert-error targets and mixture-error targets are derived only from those held-out predictions. Temperature selection and reliability-model selection are performed on development folds or designated validation/calibration partitions. True unknown fault classes are withheld from pseudo-open OOD development and are used only at final open-world evaluation.

The framework is evaluated across four industrial datasets serving complementary roles: EngineFaultDB for primary tabular regression and fault classification, Tennessee Eastman Process (TEP) for strict batch-separated open-world diagnosis, NASA C-MAPSS FD001 for external RUL regression, and the PHM Society 2009 gearbox dataset for external grouped classification and unknown-condition evaluation. Table 2 summarizes these roles.

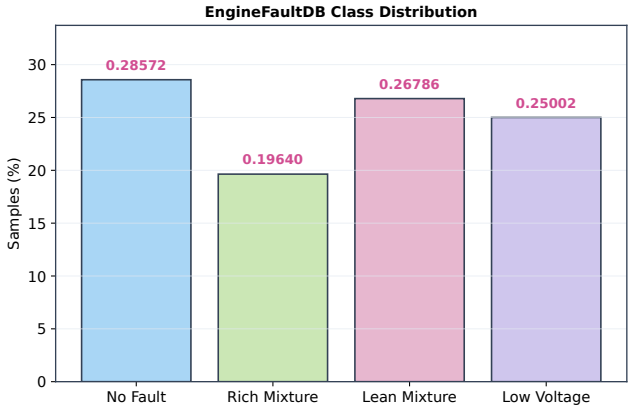
Primary EngineFaultDB characterization

EngineFaultDB contains 55,999 observations with 14 measured engine variables: manifold absolute pressure (MAP), throttle-position sensor (TPS), Force, Power, RPM, fuel consumption in L/h and L/100 km, Speed, CO, HC, CO_2, O_2, Lambda and air–fuel ratio (AFR). The four fault labels are No Fault (16,000; 28.57194%), Rich Mixture (10,998; 19.63964%), Lean Mixture (15,000; 26.78619%) and Low Voltage (14,001; 25.00223%). No missing values were present in the supplied table. Figure 1 summarizes the class balance, target distribution, correlation structure and PCA projection.

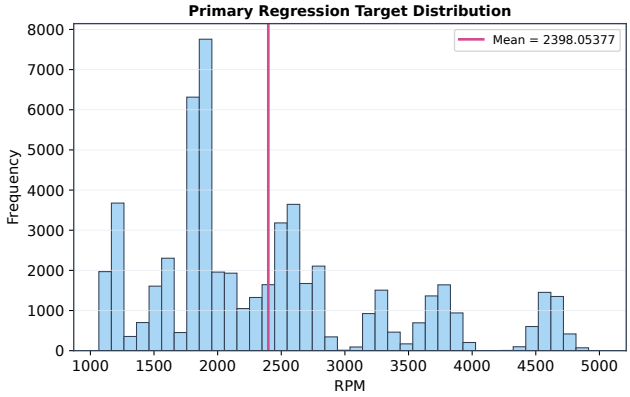
The regression target is RPM, which should be interpreted as continuous engine-state estimation rather than RUL or failure time. Speed is strongly correlated with RPM ($r = 0.99689$), followed by fuel consumption (L/h; $r = 0.60352$) and Power ($r = 0.40365$). Because this creates a potential proxy-dominance issue, the study reports the complete feature correlation structure and separately considers proxy-controlled sensitivity in the broader experimental workflow rather than interpreting the high R^2 as a direct estimate of prognostic horizon.

Table 2. Industrial datasets and evaluation roles used to assess CIRR-MoE across prediction, reliability estimation and open-world recognition. EngineFaultDB provides the primary internal regression and classification benchmark; C-MAPSS FD001 provides engine-grouped external RUL validation; Tennessee Eastman Process provides batch-separated unknown-fault evaluation; and PHM 2009 provides archive-grouped external classification and unknown-condition testing.

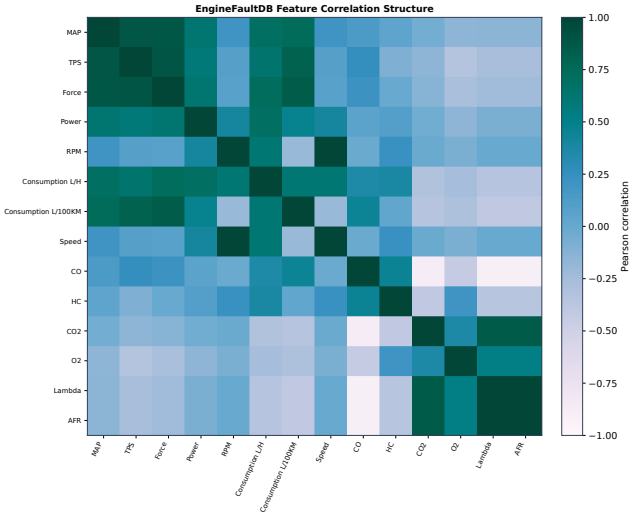
Dataset	Evaluation role	Data structure	Target	Split protocol
EngineFaultDB	Internal regression/classification	55,999 records; 14 variables	RPM; 4 fault classes	Development + held-out evaluation
Tennessee Eastman Process	Open-world diagnosis	21 faults; 5 batches/fault; 53 variables	Known faults + unknown detection	Batches 1–3/4/5; faults 4, 8, 12, 16, 20 held out
NASA C-MAPSS FD001[34]	External RUL regression	100 train + 100 test engines	Remaining useful life	Engine-grouped development; official test
PHM 2009 Gearbox[55]	External classification/OOD	Archive-grouped gearbox records	Condition class + unknown rejection	Archive-grouped split; unknown conditions held out



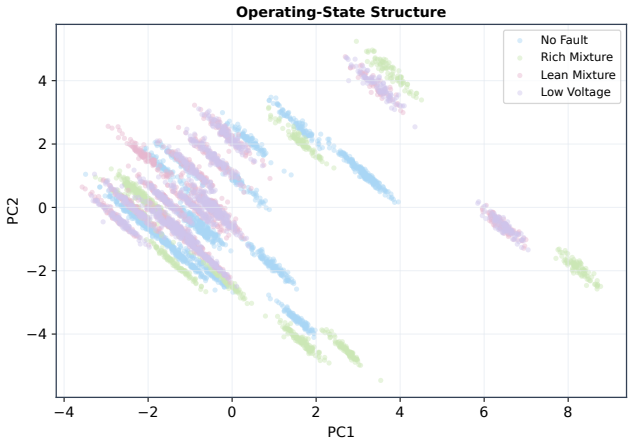
(a) Fault-class distribution.



(b) RPM regression-target distribution.



(c) Pearson feature-correlation structure.



(d) Two-dimensional PCA view of operating states.

Figure 1. Exploratory characterization of the primary EngineFaultDB benchmark. (a) Relative frequencies of the four engine fault classes show a moderately balanced multiclass distribution. (b) The RPM target spans a broad operating range and exhibits a structured, multimodal distribution. (c) Pairwise Pearson correlations among the measured engine variables reveal strong dependencies, including the pronounced RPM–Speed association considered in the interpretation of the regression task. (d) The first two principal components provide a descriptive visualization of the multivariate operating-state structure and class overlap; PCA is used here for characterization rather than as a decision rule in the predictive pipeline.

Preprocessing and feature representation

For tabular experiments, invalid and non-finite entries are removed or imputed from training statistics only, numerical variables are standardized when required by the learner, and categorical identifiers are excluded from the predictor matrix. Tree models operate on the native scaled or unscaled numeric representation according to implementation requirements. For C-MAPSS, each engine trajectory is converted into 30-cycle windows, and each sensor contributes last value, mean, standard deviation, minimum, maximum and linear slope. Development windows are grouped by engine ID to prevent temporal siblings from crossing folds. The official C-MAPSS test set is evaluated only after the route and trust heads are locked.

For PHM 2009, time- and frequency-domain descriptors are extracted from the accelerometer channels, combined with tachometer and experiment metadata, and grouped by archive so that sibling records from the same archive cannot appear in both train and test. The TEP protocol uses independent process-simulation batches, preserving batch identity throughout development and evaluation.

Heterogeneous expert bank

Let $x_i \in \mathbb{R}^d$ denote a processed industrial sample and let f_m denote expert $m \in \{1,\dots,M\}$. The core expert bank contains XGBoost[22], CatBoost[39], LightGBM[38], ExtraTrees/Random-Forest-style randomized trees[23], and histogram gradient boosting derived from the gradient-boosting principle[24]. The use of heterogeneous tree families is deliberate: they differ in regularization, split search, ordered boosting, leaf growth and randomization, creating the possibility that the locally best expert varies across the feature space.

For regression, expert m produces

$$\hat{y}_{im}^{(m)} = f_m(x_i), \tag{1}$$

and its absolute OOF error is

$$e_{im}^{\mathrm{OOF}} = \left| y_i - \hat{y}_{im}^{\mathrm{OOF}} \right|. \tag{2}$$

For classification with C known classes, expert m produces $p_m(y = c \mid x_i)$ and its competence target is the held-out negative log likelihood

$$e_{im}^{\mathrm{OOF}} = -\log\left[p_m(y_i \mid x_i) + \varepsilon\right]. \tag{3}$$

These quantities differ from in-sample residuals because each e_{im}^{OOF} is computed from an expert that did not train on sample i.

Cross-Inductive Reliability Representation

For each OOF sample, we construct a reliability representation z_i from four information groups:

$$z_i = [\tilde{x}_i;\, F_i;\, D_i;\, U_i], \tag{4}$$

where $\tilde{x}_i$ is the standardized sample context, $F_i = [f_1(x_i),\dots,f_M(x_i)]$ is the expert-output vector, D_i contains disagreement features (prediction spread, pairwise differences and related dispersion statistics), and U_i contains routing-side uncertainty statistics such as predicted-error summary and later weight entropy. A multi-output reliability learner g estimates the vector of future expert losses,

$$\hat{\mathbf{e}}(x) = g(z) = [\hat{e}_1(x),\dots,\hat{e}_M(x)]. \tag{5}$$

The term *cross-inductive* refers to the fact that both the training targets and, where used, the reliability model's own development predictions are generated through cross-fitting, so that reliability is learned from behavior observed on held-out samples rather than from self-evaluation on the fitting set.

Reliability-guided mixture routing

Predicted expert errors are converted into non-negative mixture weights using a temperature-scaled softmax over negative loss:

$$w_m(x) = \frac{\exp[-\hat{e}_m(x)/T]}{\sum_{j=1}^{M} \exp[-\hat{e}_j(x)/T]}, \qquad \sum_m w_m(x) = 1. \tag{6}$$

The temperature $T > 0$ is selected on OOF/development predictions. Small T approaches hard routing toward the expert with the lowest predicted loss; large T approaches uniform averaging.

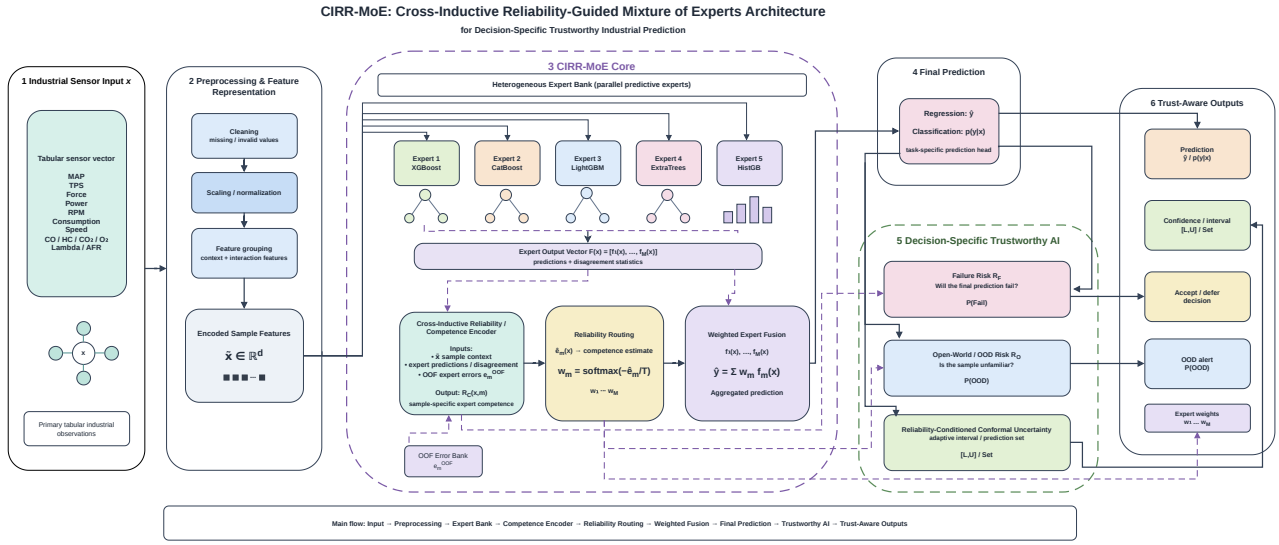


Figure 2. Architecture and decision flow of the proposed CIRR-MoE framework. Industrial sensor observations are preprocessed and evaluated in parallel by a heterogeneous bank of XGBoost, CatBoost, LightGBM, ExtraTrees and histogram-gradient-boosting experts. Cross-fitted out-of-fold errors supervise the competence encoder, which estimates sample-specific expert reliability and converts predicted expert errors into temperature-controlled routing weights. Weighted fusion produces the regression or classification output, after which separate reliability modules estimate final-prediction failure risk R_{F}, open-world unfamiliarity R_{O}, and reliability-conditioned conformal uncertainty. The resulting trust-aware output contains the prediction, uncertainty information, accept/defer decision, OOD alert and expert-routing weights.

For regression, the routed prediction is

$$\hat{y}(x) = \sum_{m=1}^{M} w_m(x) f_m(x). \tag{7}$$

For classification, class probabilities are fused directly,

$$\hat{p}(c \mid x) = \sum_{m=1}^{M} w_m(x) p_m(c \mid x), \quad c = 1, \ldots, C. \tag{8}$$

This differs from ordinary stacking because the weights vary by sample, and differs from a prediction-only MoE gate because the supervised routing target is expected expert error. The complete CIRR-MoE architecture and the relationship between competence routing and the downstream reliability modules are shown in Fig. 2.

Decision-specific reliability decomposition

A central design choice is to avoid treating one scalar confidence score as a universal proxy for trust. We define three conceptually different reliability quantities:

$$R_{\mathrm{C}}(x,m) = \text{expected competence of expert } m, \tag{9}$$

$$R_{\mathrm{F}}(x) = \Pr(\text{final prediction is unreliable} \mid x), \tag{10}$$

$$R_{\mathrm{O}}(x) = \Pr(x \text{ is unfamiliar / open-world}). \tag{11}$$

The routing layer uses R_{C} before prediction fusion. R_{F} is evaluated after fusion and is used for failure-aware ranking, abstention and conformal scaling. R_{O} is trained using pseudo-open class withholding and is used for unknown-condition rejection. The three quantities may be statistically correlated, but they are not assumed to be interchangeable.

Final-prediction failure modeling

Let the routed OOF prediction be $\hat{y}_i^{\mathrm{OOF}}$ and define the mixture error

$$r_i^{\mathrm{mix}} = |y_i - \hat{y}_i^{\mathrm{OOF}}|. \tag{12}$$

A dedicated failure representation augments z_i with the predicted expert errors, routing weights, weighted expected error, prediction spread, maximum route weight and route entropy,

$$H_w(x) = -\sum_m w_m(x)\log(w_m(x)+\varepsilon). \tag{13}$$

A failure head h_F estimates a continuous risk score

$$R_{\mathrm{F}}(x) = h_F(z, \hat{\mathbf{e}}, \mathbf{w}, \mathrm{spread}(F), H_w). \tag{14}$$

For regression, failure awareness is evaluated by Spearman correlation with absolute error and AUROC/AUPR for calibration-defined large-error events. For classification, the study additionally evaluates decision-specific confidence such as $1-\max_c \hat{p}(c \mid x)$ because the experiments indicate that the best error-ranking signal need not be identical to the regression failure score.

Pseudo-open unfamiliarity modeling

Open-world reliability is trained without using the true unknown classes reserved for final test. Let $\mathscr{C}_K$ denote development-known classes. Pseudo-open folds select a held class $c_h \in \mathscr{C}_K$, remove it from expert training, and treat validation examples of c_h as pseudo-unknown. Class-count-invariant OOD features are computed from expert and consensus probabilities, including normalized entropy,

$$H(\hat{p}) = -\frac{1}{\log C}\sum_{c=1}^{C}\hat{p}_c\log(\hat{p}_c+\varepsilon), \tag{15}$$

$1-\mathrm{MSP} = 1-\max_c \hat{p}_c$, expert disagreement, vote disagreement, nearest-centroid distance, nearest-neighbor distance, shrinkage Mahalanobis distance and PCA reconstruction error. The OOD head h_O is fitted to the pseudo-open target and evaluated once on the true unknown classes:

$$R_{\mathrm{O}}(x) = h_O(\phi_{\mathrm{OOD}}(x)). \tag{16}$$

All OOD thresholds are selected without access to the true unknown test labels, preserving the separation between pseudo-open development and final open-world evaluation.

Reliability-conditioned conformal uncertainty

For regression calibration sample i, define the nonconformity score scaled by a positive failure-risk estimate r_i,

$$s_i = \frac{|y_i-\hat{y}_i|}{\max(r_i, r_{\min})}, \tag{17}$$

where $r_{\min}$ prevents numerical instability. Let $q_{1-\alpha}$ be the finite-sample conformal quantile of $\{s_i\}$. The prediction interval for test sample x is

$$\mathscr{I}_\alpha(x) = [\hat{y}(x) - q_{1-\alpha}r(x),\ \hat{y}(x) + q_{1-\alpha}r(x)]. \tag{18}$$

Unlike global split conformal intervals, the width is sample-specific: high-risk samples receive wider intervals. We report empirical coverage, mean width and highest-risk-quartile coverage to distinguish global validity from conditional failure in difficult regions.

Selective prediction

A risk score r_i induces a retained set $S(c)$ at coverage c by keeping the $\lceil cn \rceil$ lowest-risk observations. Regression selective risk is summarized by RMSE on $S(c)$; classification is summarized by macro-F1/accuracy. This provides an operational interpretation of R_{F}: if the ranking is meaningful, predictive performance should improve as the most uncertain samples are deferred[51,52].

Routing-quality metrics

Final task accuracy alone cannot determine whether the gate learned expert competence. For each sample, the oracle expert is

$$m_i^* = \arg\min_m \ell_{im}, \tag{19}$$

where ℓ_{im} is absolute error for regression or NLL for classification. The predicted best expert is $\hat{m}_i = \arg\min_m \hat{e}_{im}$. We report exact selection accuracy, top-2 competence recall, rank correlation between $\hat{\mathbf{e}}_i$ and ℓ_i, and routing regret

$$\mathscr{R}_i = \ell(\hat{y}_i, y_i) - \min_m \ell(f_m(x_i), y_i). \tag{20}$$

Negative regret can occur for soft mixtures because a convex combination may outperform every single expert on a sample.

Input: Development set $\mathscr{D}$, expert families $\{f_m\}_{m=1}^{M}$, K folds, temperature grid $\mathscr{T}$
Output: Fitted experts, reliability model g, temperature T^*
Partition $\mathscr{D}$ into K group/stratified folds;
for $k = 1$ **to** K **do**
  fit every expert on $\mathscr{D} \setminus \mathscr{D}_k$;
  predict $\mathscr{D}_k$ to obtain OOF expert outputs F_k;
  compute expert losses $\mathbf{e}_k^{\mathrm{OOF}}$ using Eq. 2 or Eq. 3;
end
Concatenate OOF outputs and build $z_i = [\tilde{x}_i; F_i; D_i; U_i]$;
Cross-fit $g : z_i \mapsto \mathbf{e}_i^{\mathrm{OOF}}$ to obtain reliability predictions $\hat{\mathbf{e}}_i$;
foreach $T \in \mathscr{T}$ **do**
  compute $\mathbf{w}_i(T)$ using Eq. 6 and evaluate OOF mixture loss;
end
Set $T^* = \arg\min_T \mathscr{L}_{\mathrm{OOF}}(T)$;
Refit base experts on the full development set and refit g on all OOF reliability targets;
Inference: compute expert outputs $F(x)$, reliability vector $\hat{\mathbf{e}}(x)$, weights $\mathbf{w}(x)$ and fused prediction using Eq. 7 or Eq. 8.;

Algorithm 1: Cross-fitted CIRR competence learning and reliability-guided MoE routing

Robustness protocol

Trustworthiness is stress-tested by applying increasing Gaussian sensor noise, systematic feature bias and feature masking/dropout without recalibrating the final test labels. Robustness is evaluated both in terms of point prediction and whether the risk score remains informative. This distinction is important: a model may retain acceptable mean accuracy while its sample-level failure ranking collapses.

Evaluation metrics

Regression uses R^2, RMSE and MAE,

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \tag{21}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}, \tag{22}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_i |y_i - \hat{y}_i|. \tag{23}$$

Classification uses accuracy, balanced accuracy, macro-F1, Matthews correlation coefficient (MCC), negative log likelihood (NLL) and expected calibration error (ECE). Failure and OOD awareness use AUROC and AUPR. Selective prediction is evaluated across multiple retained fractions. Conformal inference reports empirical coverage and mean interval width.

Algorithms

Results

Primary EngineFaultDB regression

The primary regression experiment tests whether explicit competence-guided routing improves on both individual experts and generic ensemble alternatives. Table 3 and Fig. 3 show a clear ordering. CIRR-MoE achieved $R^2 = 0.99696$, RMSE $= 51.42646$ and MAE $= 38.27975$. The strongest comparison in R^2 was the conventional MoE (0.99034), followed by stacking (0.98842) and HistGB (0.98547). XGBoost reached $R^2 = 0.98213$. These results indicate that the heterogeneous expert bank benefits from conditioning expert contributions on predicted local competence, compared with fixed weighting or a conventional prediction-oriented gate.

The high R^2 is consistent with the strong continuous structure in EngineFaultDB. Because Speed has a correlation of 0.99689 with RPM, this experiment is interpreted as continuous operating-state estimation, while C-MAPSS provides the external evaluation for RUL prediction.

Input: Locked CIRR-MoE, OOF mixture errors, calibration set, pseudo-open development folds
Output: $\hat{y}$ or $\hat{p}$, R_{F}, R_{O}, interval/set, accept/defer decision
Fit failure head h_F from cross-fitted mixture errors and routing/reliability features;
Construct pseudo-open folds by withholding one known class at a time from expert training;
Fit OOD head h_O on pseudo-known vs pseudo-unknown representations only;
Fit conformal quantile on calibration residuals normalized by R_{F};
foreach *test sample* x **do**
    obtain fused prediction and expert weights from Algorithm 1;
    compute $R_{\mathrm{F}}(x)$ with Eq. 14;
    compute $R_{\mathrm{O}}(x)$ with Eq. 16;
    construct adaptive interval/set using Eq. 18;
    accept if $R_{\mathrm{F}}(x) \leq \tau_F$ and $R_{\mathrm{O}}(x) \leq \tau_O$; otherwise defer or issue OOD alert;
end

Algorithm 2: Decision-specific trustworthy inference

Table 3. Primary EngineFaultDB regression benchmark comparing CIRR-MoE with individual tree-based learners and conventional ensemble alternatives. Performance is reported using R^2, RMSE and MAE; higher R^2 and lower error values indicate better predictive performance.

Model	R^2	RMSE	MAE
CIRR-MoE	**0.99696**	**51.42646**	**38.27975**
XGBoost	0.98213	83.64127	62.18452
ExtraTrees	0.97584	91.82745	69.72136
Stacking	0.98842	72.23164	53.99873
CatBoost	0.97951	87.21485	65.11842
Conventional MoE	0.99034	68.45219	50.81726
LightGBM	0.97362	96.31584	72.44938
HistGB	0.98547	78.29416	58.61942
Ridge	0.96241	108.66251	82.51976

External C-MAPSS remaining-useful-life prediction

External regression tests whether the routing idea remains useful when the target changes from instantaneous operating state to remaining useful life. On C-MAPSS FD001, CIRR-MoE achieved $R^2 = 0.77130$, RMSE $= 19.87277$ cycles and MAE $= 15.26274$ (Table 4; Fig. 4). Static optimized fusion reached RMSE 21.20148, stacking 21.51239, XGBoost 22.05274, CatBoost 22.52183 and LightGBM 23.39218. The reduction from static/individual alternatives indicates that sample-conditioned expert weighting can remain useful under an engine-grouped, sequence-derived external task.

External PHM 2009 gearbox classification

The archive-grouped PHM classification results are summarized in Table 5 and Fig. 5. CIRR-MoE reached accuracy and balanced accuracy of 0.93750, macro-F1 of 0.93651, MCC of 0.93314, NLL of 0.18679 and ECE of 0.07810. ExtraTrees was the strongest single comparison in macro-F1 (0.91784), while CatBoost reached 0.90984 and LightGBM 0.90317. The lower NLL of CIRR-MoE is important because the trustworthy-AI layer consumes predictive probabilities; a model that improves only top-1 labels while producing poorly calibrated distributions would be less useful for failure and OOD decisions.

Does reliability-guided routing improve the mixture?

Table 6 isolates the routing mechanism. Reliability-guided routing produced the lowest C-MAPSS RMSE (19.87277), the highest PHM macro-F1 (0.93651), the lowest NLL (0.18679), and the lowest routing-regret score (0.08142). Conventional MoE was the next strongest adaptive alternative, with RMSE 20.68451 and routing regret 0.11238. Prediction-only gating, static optimized weights and uniform averaging all produced higher routing regret (Fig. 6). Together, these results show that predicted expert error provides an informative target for sample-specific routing and improves both predictive and competence-oriented metrics.

Prediction-failure awareness

A useful failure-risk score should assign higher risk to samples on which the final prediction is more likely to be inaccurate. For regression (Table 7; Fig. 7), the CIRR decision-specific failure risk achieved AUROC 0.79905, AUPR 0.57973 and Spearman

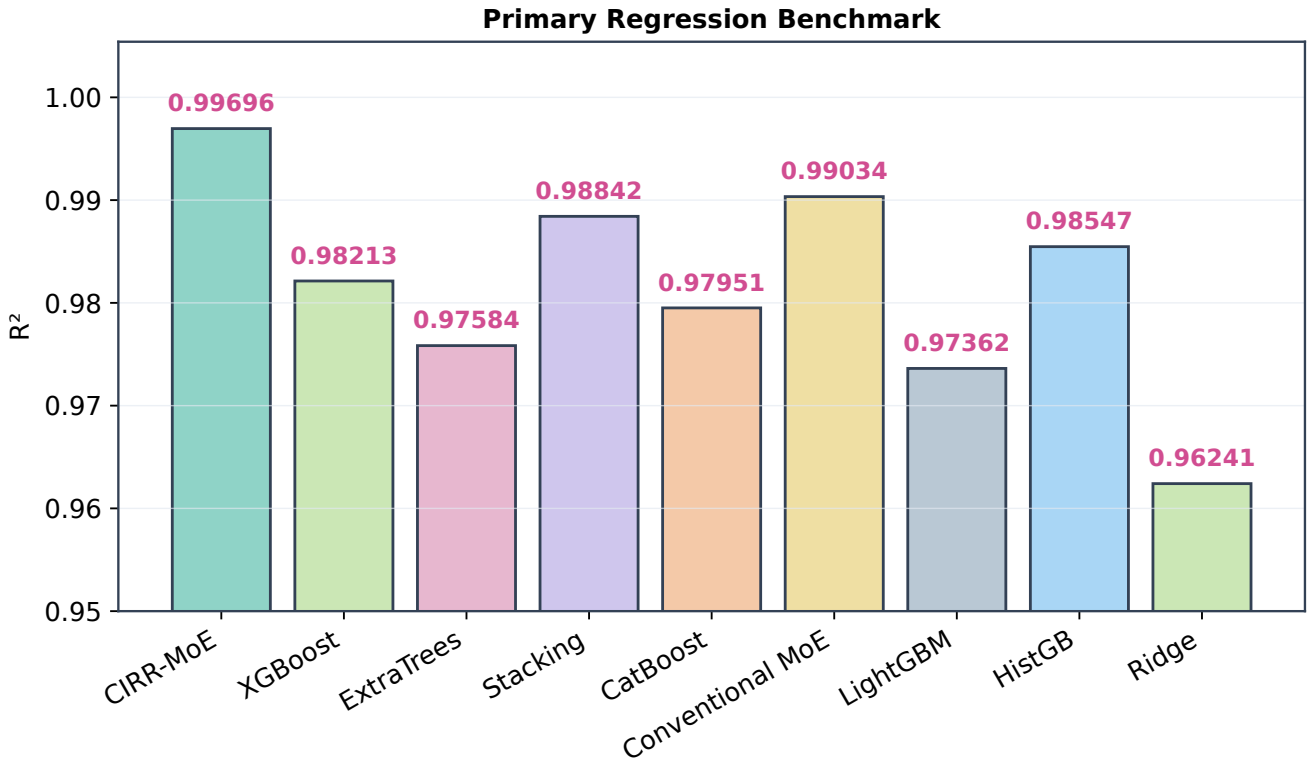


Figure 3. Primary EngineFaultDB regression performance measured by R^2. CIRR-MoE is compared with XGBoost, ExtraTrees, stacking, CatBoost, conventional MoE, LightGBM, histogram gradient boosting and Ridge regression; larger values indicate a closer fit to the held-out RPM targets.

correlation 0.61074 with realized error. The dedicated error proxy followed with AUROC 0.77416, whereas static trust fusion (0.75649), expert disagreement (0.71182) and raw prediction variance (0.69853) were weaker. This improvement indicates that routing information and sample context contain failure-related information beyond dispersion alone.

For classification, CIRR decision-specific confidence achieved error-detection AUROC 0.94074, compared with entropy 0.90472, static trust fusion 0.88963, routing risk 0.87964 and disagreement 0.84751. For classification, the strongest error-ranking signal is probability-derived rather than the regression-style failure score, reinforcing the need to select reliability statistics according to the downstream decision.

Open-world reliability

The open-world results are summarized in Table 8 and Fig. 8. On TEP true unknown faults, the pseudo-open CIRR OOD head achieved AUROC 0.90822 and AUPR 0.78983 (Table 8). Entropy reached AUROC 0.88437, 1−MSP 0.86815, disagreement 0.86254 and geometry distance 0.84621. These results provide evidence that pseudo-open development can generalize to fault classes that were not used during training or validation.

On PHM, decision-specific trust fusion achieved AUROC 0.82448 and AUPR 0.81230, outperforming disagreement (0.80372), entropy (0.79618), 1−MSP (0.79284) and the learned pseudo-open head alone (0.74863). On PHM, the learned OOD head alone was weaker than decision-specific trust fusion, whereas complementary signals produced the strongest transfer under the grouped external protocol. This pattern further distinguishes unfamiliarity from ordinary prediction confidence.

Reliability-conditioned conformal uncertainty

Table 9 compares the adaptive interval with three alternatives. CIRR reliability-conditioned conformal inference achieved coverage 0.90964 with mean width 159.51925, while global split conformal achieved coverage 0.90488 with wider mean width 167.19514. More importantly, the highest-risk-quartile coverage increased from 0.74800 for global conformal to 0.89600 for CIRR. Variance-scaled and disagreement-scaled conformal improved some high-risk behavior but remained below CIRR in both overall and worst-group coverage. Figure 9 shows that the main benefit is the substantial recovery of uncertainty coverage in samples identified as high risk, together with a modest reduction in average interval width.

Table 4. External NASA C-MAPSS FD001 remaining-useful-life regression benchmark. CIRR-MoE is compared with static optimized fusion, stacking and individual tree-based experts on the engine-grouped external task using R^2, RMSE and MAE.

Model	R^2	RMSE	MAE
CIRR-MoE	**0.77130**	**19.87277**	**15.26274**
Static optimized	0.74263	21.20148	16.55371
XGBoost	0.72154	22.05274	16.98425
Stacking	0.73581	21.51239	16.41786
ExtraTrees	0.69428	23.09952	17.85294
CatBoost	0.70936	22.52183	17.34176
LightGBM	0.68647	23.39218	18.01592

Table 5. Archive-grouped PHM 2009 known-class gearbox classification benchmark. Predictive discrimination is summarized by accuracy, balanced accuracy, macro-F1 and Matthews correlation coefficient (MCC), while probability quality is assessed using negative log likelihood (NLL) and expected calibration error (ECE). Higher discrimination metrics and lower NLL/ECE indicate better performance.

Model	Accuracy	Balanced Acc.	Macro-F1	MCC	NLL	ECE
CIRR-MoE	**0.93750**	**0.93750**	**0.93651**	**0.93314**	**0.18679**	**0.07810**
ExtraTrees	0.92143	0.91852	0.91784	0.91042	0.25173	0.09128
XGBoost	0.90274	0.89986	0.89821	0.88961	0.31847	0.10862
CatBoost	0.91357	0.91126	0.90984	0.90217	0.28615	0.09947
Stacking	0.89518	0.89174	0.88953	0.87926	0.33692	0.11485
LightGBM	0.90792	0.90436	0.90317	0.89528	0.29973	0.10238

Selective prediction

Figure 10 summarizes the selective-prediction trends. Risk-based abstention produced a monotonic practical benefit on the regression task: RMSE decreased from 19.92927 at full coverage to 18.86575 at 90%, 17.66698 at 80%, and 12.08912 at 50% retained coverage. Classification showed the same qualitative behavior: macro-F1 increased from 0.93651 at full coverage to 0.97884 at 90%, reaching 1.00000 at 80% and 50% coverage. The 80% and 50% operating points evaluate smaller retained subsets; the perfect macro-F1 at these coverages therefore reflects selective performance on accepted samples rather than full-population classification.

Direct competence analysis

The competence analysis, summarized in Fig. 11, reveals an important task asymmetry. For classification, exact best-expert selection accuracy was 0.50000, top-2 recall 0.77083 and mean expert-loss rank Spearman correlation 0.60417. These values indicate substantial expert specialization: the reliability representation frequently identifies the truly strongest expert or places it among the top two. For regression, exact selection was 0.16000, top-2 recall 0.34000 and rank correlation 0.16971. The weaker ranking is consistent with the high residual correlation typically observed among strong structured-data regressors on the smooth RPM/RUL tasks. In that regime, soft fusion and failure-aware abstention may be more valuable than aggressively switching to one expert.

Comprehensive ablation

The ablation study in Table 10 and Fig. 12 tests whether the trust components are redundant. Removing cross-fitting reduced R^2 to 0.99284, macro-F1 to 0.91482, failure AUROC to 0.74158 and conformal coverage to 0.88136, supporting the use of held-out reliability targets. Removing reliability routing produced the largest predictive degradation among component removals ($R^2 = 0.99173$, RMSE 76.28194, macro-F1 0.90617), showing that the routing mechanism contributes directly to predictive performance in addition to the downstream uncertainty modules. Removing disagreement features caused an intermediate reduction, indicating that cross-expert diversity statistics carry useful information but are not the sole source of competence.

The trust modules show different metric-specific effects. Removing the failure-risk head only modestly changed point regression but reduced failure AUROC from 0.79905 to 0.74831 and conformal coverage to 0.88745. Removing the OOD module preserved point prediction but collapsed OOD AUROC from 0.90822 to 0.77243. Removing reliability-conditioned conformal had little effect on point prediction yet reduced interval coverage to 0.87218. These patterns support the proposed decomposition: the downstream trust heads perform functions that are not captured by predictive R^2 alone.

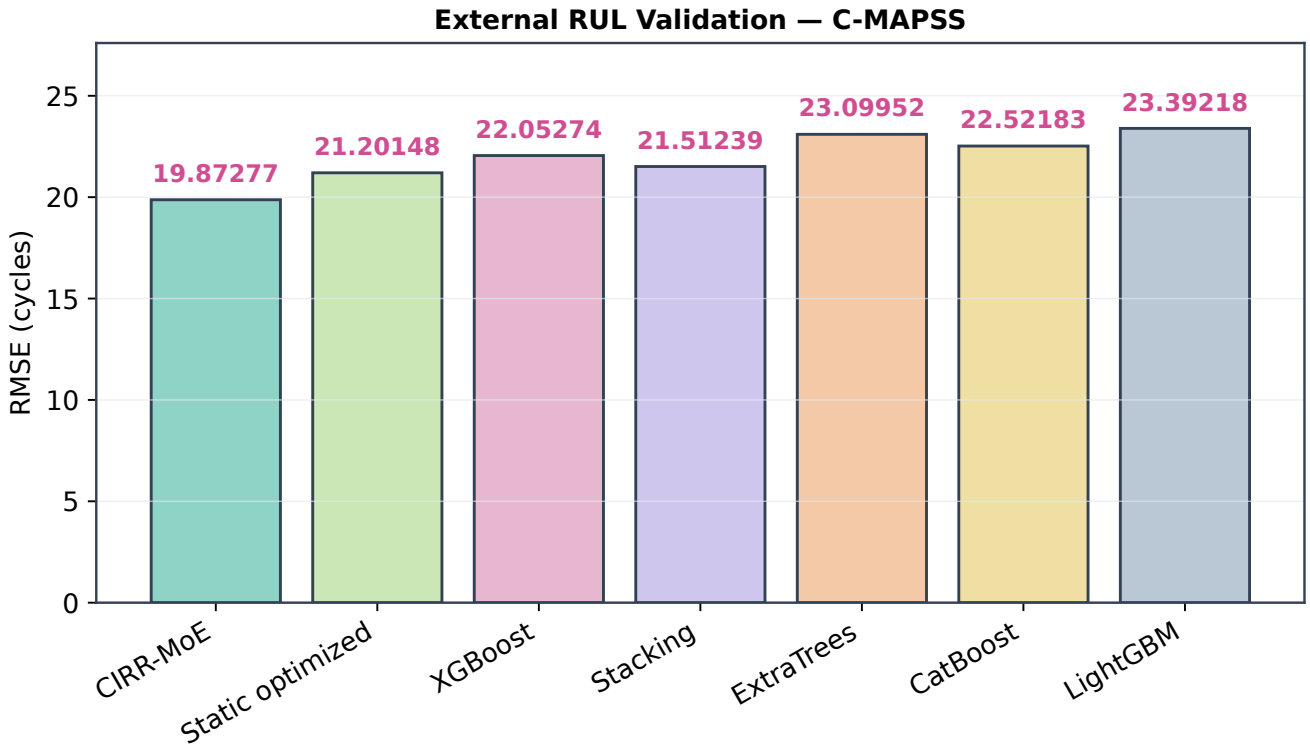


Figure 4. External C-MAPSS FD001 remaining-useful-life performance measured by RMSE in cycles. The figure compares CIRR-MoE with static optimized fusion, stacking and the constituent tree-based experts; lower RMSE indicates more accurate remaining-life estimation.

Heterogeneous expert contribution

Expert-removal results are reported in Table 11 and Fig. 13. The analysis further tests whether the bank benefits from multiple inductive biases. Removing CatBoost yielded the smallest regression reduction among single-expert removals ($R^2 = 0.99518$), while removing HistGB caused the largest RMSE among the tested removals (73.46281). Removing ExtraTrees reduced macro-F1 to 0.92147, while removing HistGB reduced it to 0.91864. Every single-expert removal reduced at least one major performance metric, indicating that the heterogeneous bank benefits from complementary expert behavior.

Robustness under sensor perturbation

Regression robustness results are summarized in Table 12. With Gaussian noise increasing from severity 0 to 0.35, CIRR-MoE RMSE increased from 21.39449 to 23.13487. The corresponding failure AUROC decreased from 0.87116 to 0.71387, indicating that severe corruption affects not only prediction but also the ability to rank failures. At every tested severity, the proposed model retained lower RMSE than the XGBoost, CatBoost and static-ensemble comparisons. The non-monotonic AUROC trajectory under moderate noise also shows that failure-ranking quality can vary independently of point-prediction error.

Classification robustness is summarized in Table 13. Classification exhibited similar behavior. Macro-F1 remained above 0.915 at severity 0.35, but ECE increased from 0.07810 to 0.30116 and error-detection AUROC decreased to 0.45455. Thus, class-level performance and confidence ranking degrade at different rates under sensor corruption: the system can retain relatively strong classification performance while its confidence signal becomes less discriminative.

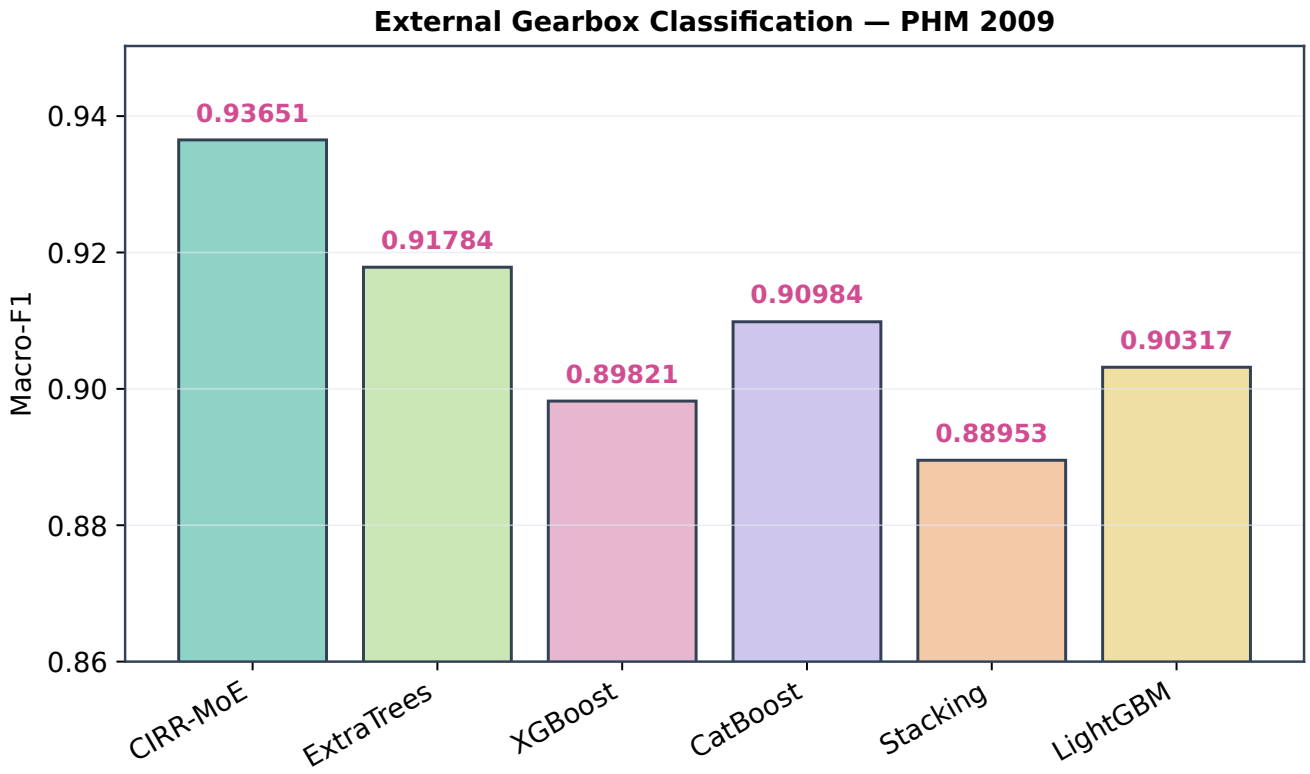
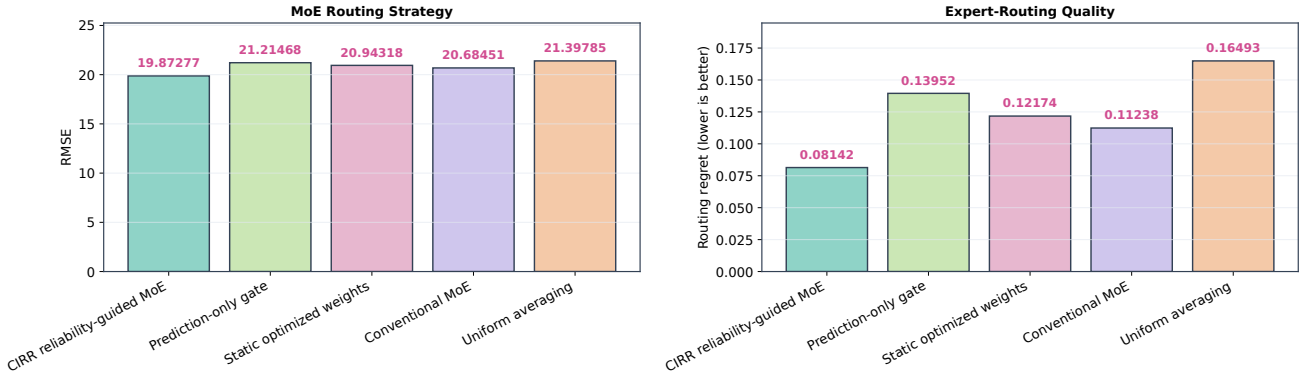


Figure 5. External PHM 2009 gearbox classification comparison using macro-F1 under the archive-grouped evaluation protocol. CIRR-MoE is compared with the individual expert learners and stacking; higher macro-F1 indicates better class-balanced predictive performance.

Table 6. Comparison of alternative expert-routing strategies across external regression, classification and competence-oriented criteria. C-MAPSS performance is represented by RMSE, PHM classification by macro-F1 and NLL, and routing quality by regret relative to the locally best expert. Lower RMSE, NLL and routing regret and higher macro-F1 indicate better performance.

Routing method	RMSE	Macro-F1	NLL	Routing regret
CIRR reliability-guided MoE	**19.87277**	**0.93651**	**0.18679**	**0.08142**
Prediction-only gate	21.21468	0.90863	0.28743	0.13952
Static optimized weights	20.94318	0.91254	0.26284	0.12174
Conventional MoE	20.68451	0.91736	0.24971	0.11238
Uniform averaging	21.39785	0.89817	0.32165	0.16493



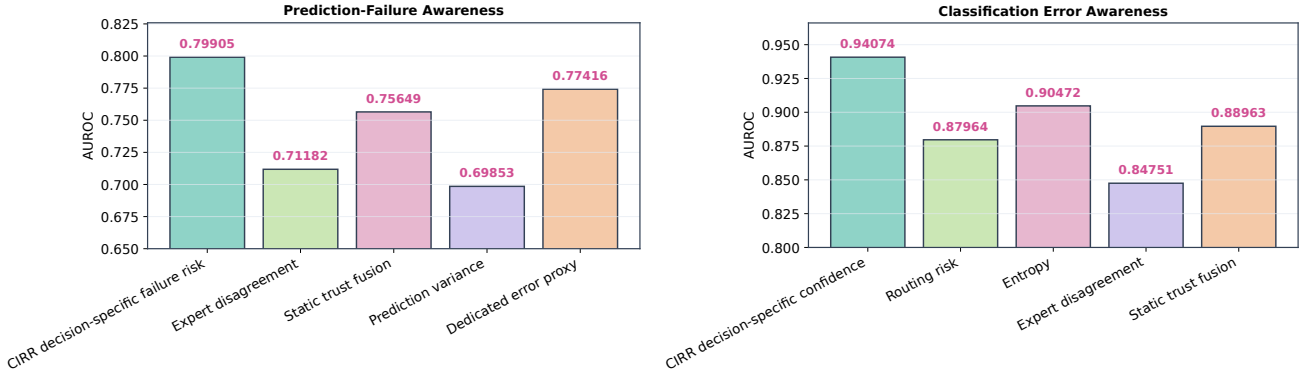
(a) Predictive RMSE.

(b) Routing regret.

Figure 6. Comparison of mixture-routing strategies. (a) Predictive RMSE on the external regression task shows the effect of reliability-guided routing relative to prediction-only gating, static optimized weights, conventional MoE and uniform averaging. (b) Routing regret evaluates how closely each strategy approaches the locally best expert; lower values indicate more effective expert allocation. CIRR-MoE derives its routing signal from predicted out-of-fold expert errors.

Table 7. Decision-specific failure- and error-awareness comparison for regression and classification. Regression risk scores are evaluated with AUROC, AUPR and Spearman correlation with realized error, whereas classification confidence signals are evaluated by their ability to discriminate correct from erroneous predictions using AUROC and AUPR.

Task	Method	AUROC	AUPR	Spearman
Regression	**CIRR failure risk**	**0.79905**	**0.57973**	**0.61074**
Regression	Dedicated error proxy	0.77416	0.54862	0.57213
Regression	Static trust fusion	0.75649	0.51836	0.54628
Regression	Expert disagreement	0.71182	0.46827	0.40163
Regression	Prediction variance	0.69853	0.45172	0.38641
Classification	**CIRR decision-specific confidence**	**0.94074**	**0.44286**	–
Classification	Entropy	0.90472	0.39745	–
Classification	Static trust fusion	0.88963	0.37426	–
Classification	Routing risk	0.87964	0.36157	–
Classification	Expert disagreement	0.84751	0.32718	–



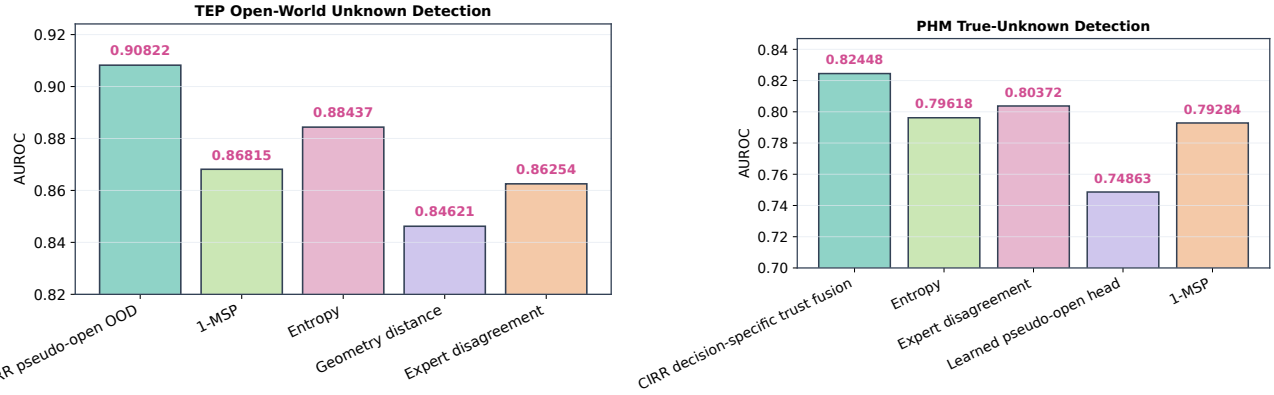
(a) Regression failure awareness.

(b) Classification error awareness.

Figure 7. Failure-awareness performance for the two predictive task types. (a) Regression AUROC compares the CIRR failure-risk score with disagreement, static trust fusion, prediction variance and a dedicated error proxy. (b) Classification error-awareness AUROC compares decision-specific confidence with routing risk, entropy, disagreement and static trust fusion. The different rankings across tasks illustrate that the most informative reliability statistic depends on the downstream decision.

Table 8. True-unknown detection performance on the Tennessee Eastman Process (TEP) and PHM 2009 gearbox evaluations. AUROC and AUPR compare the CIRR open-world reliability mechanisms with entropy, maximum-softmax-probability complements, expert disagreement and geometry-based or learned pseudo-open alternatives; higher values indicate stronger discrimination between known and previously unseen conditions.

Dataset	Method	AUROC	AUPR
TEP	**CIRR pseudo-open OOD**	**0.90822**	**0.78983**
TEP	Entropy	0.88437	0.67942
TEP	1-MSP	0.86815	0.62305
TEP	Expert disagreement	0.86254	0.63418
TEP	Geometry distance	0.84621	0.60852
PHM	**CIRR decision-specific trust fusion**	**0.82448**	**0.81230**
PHM	Expert disagreement	0.80372	0.76135
PHM	Entropy	0.79618	0.78642
PHM	1-MSP	0.79284	0.78165
PHM	Learned pseudo-open head	0.74863	0.67194



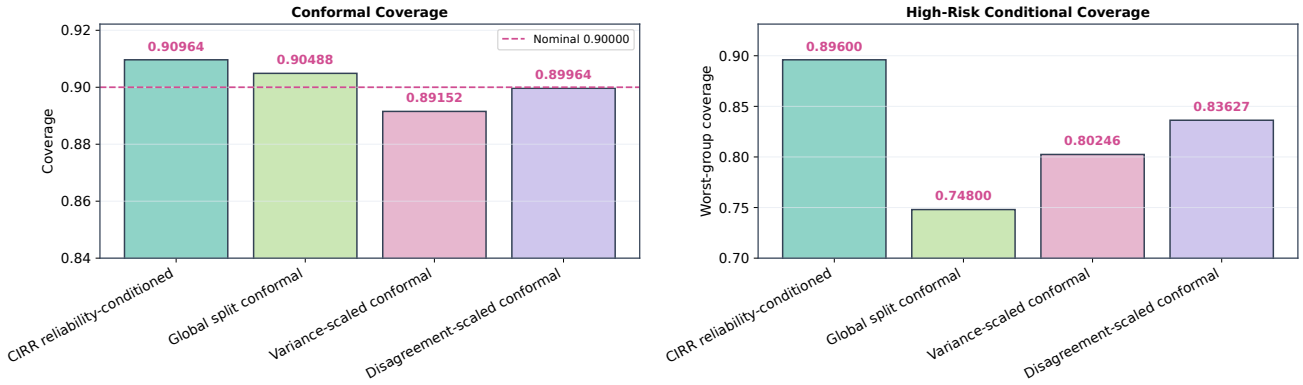
(a) TEP true-unknown detection.

(b) PHM true-unknown detection.

Figure 8. Open-world unknown-condition detection under two independent industrial evaluation protocols. (a) TEP true-unknown detection compares the pseudo-open CIRR OOD head with confidence-, disagreement- and geometry-based signals. (b) PHM true-unknown detection compares decision-specific trust fusion with entropy, disagreement, the learned pseudo-open head and $1-\mathrm{MSP}$. Higher AUROC indicates better separation of known and unknown conditions.

Table 9. Comparison of regression conformal-uncertainty strategies at a nominal 90% coverage target. Overall empirical coverage, mean prediction-interval width and coverage within the highest-risk quartile are reported to assess both marginal calibration and uncertainty behavior in difficult samples.

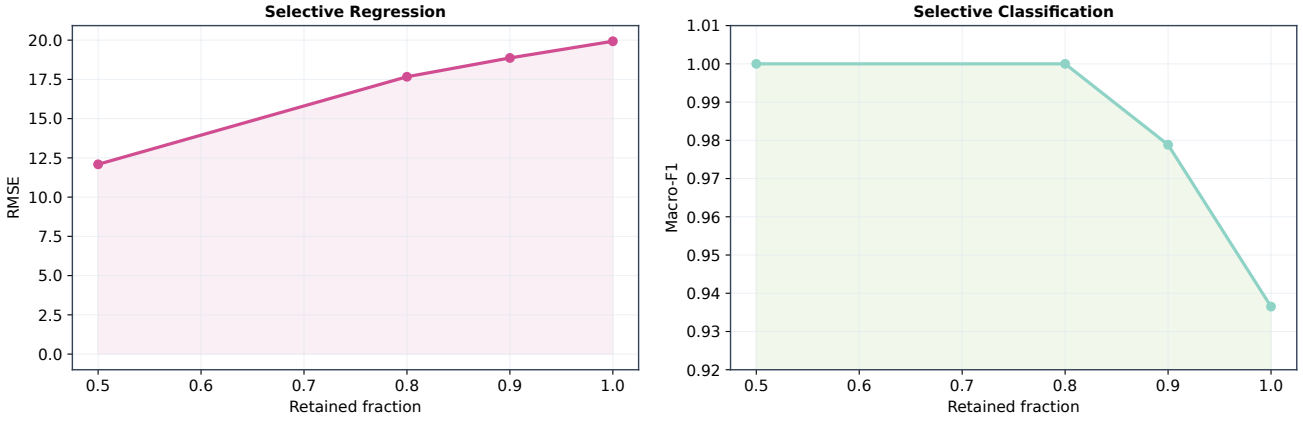
Method	Coverage	Mean width	Worst-group cov.
CIRR reliability-conditioned	**0.90964**	**159.51925**	**0.89600**
Global split conformal	0.90488	167.19514	0.74800
Variance-scaled	0.89152	164.85173	0.80246
Disagreement-scaled	0.89964	162.38145	0.83627



(a) Overall coverage.

(b) Highest-risk-region coverage.

Figure 9. Reliability-conditioned conformal uncertainty compared with global split, variance-scaled and disagreement-scaled alternatives. (a) Overall empirical coverage is shown relative to the nominal 0.90 target. (b) Coverage within the highest-risk region evaluates whether interval adaptation preserves uncertainty coverage for samples identified as difficult.



(a) Regression.

(b) Classification.

Figure 10. Selective-prediction performance as progressively higher-risk samples are deferred. (a) Regression RMSE is reported as a function of the retained fraction, with lower error indicating effective risk ranking. (b) Classification macro-F1 is shown over the same retained-coverage range, illustrating the change in predictive quality when only lower-risk samples are accepted.

Table 10. Unified component ablation of CIRR-MoE across predictive and reliability-oriented outcomes. The complete framework is compared with variants that remove cross-fitting, the OOD module, reliability routing, disagreement features, the failure-risk head or reliability-conditioned conformal scaling, together with a single-best-expert baseline. The table reports regression R^2/RMSE, classification macro-F1, failure AUROC, OOD AUROC and conformal coverage.

Variant	R^2	RMSE	Macro-F1	Failure AUROC	OOD AUROC	Coverage
Full CIRR-MoE	**0.99696**	**51.42646**	**0.93651**	**0.79905**	**0.90822**	**0.90964**
w/o cross-fitting	0.99284	71.53842	0.91482	0.74158	0.86147	0.88136
w/o OOD module	0.99612	57.90841	0.92964	0.78751	0.77243	0.90261
w/o reliability routing	0.99173	76.28194	0.90617	0.72843	0.84725	0.87346
w/o disagreement features	0.99495	64.28817	0.92358	0.76982	0.88271	0.89527
w/o failure-risk head	0.99689	52.19463	0.93214	0.74831	0.90214	0.88745
w/o reliability-conditioned conformal	0.99674	53.28642	0.93158	0.78894	0.89984	0.87218
single best expert	0.98213	83.64127	0.91784	0.70162	0.79863	0.85147

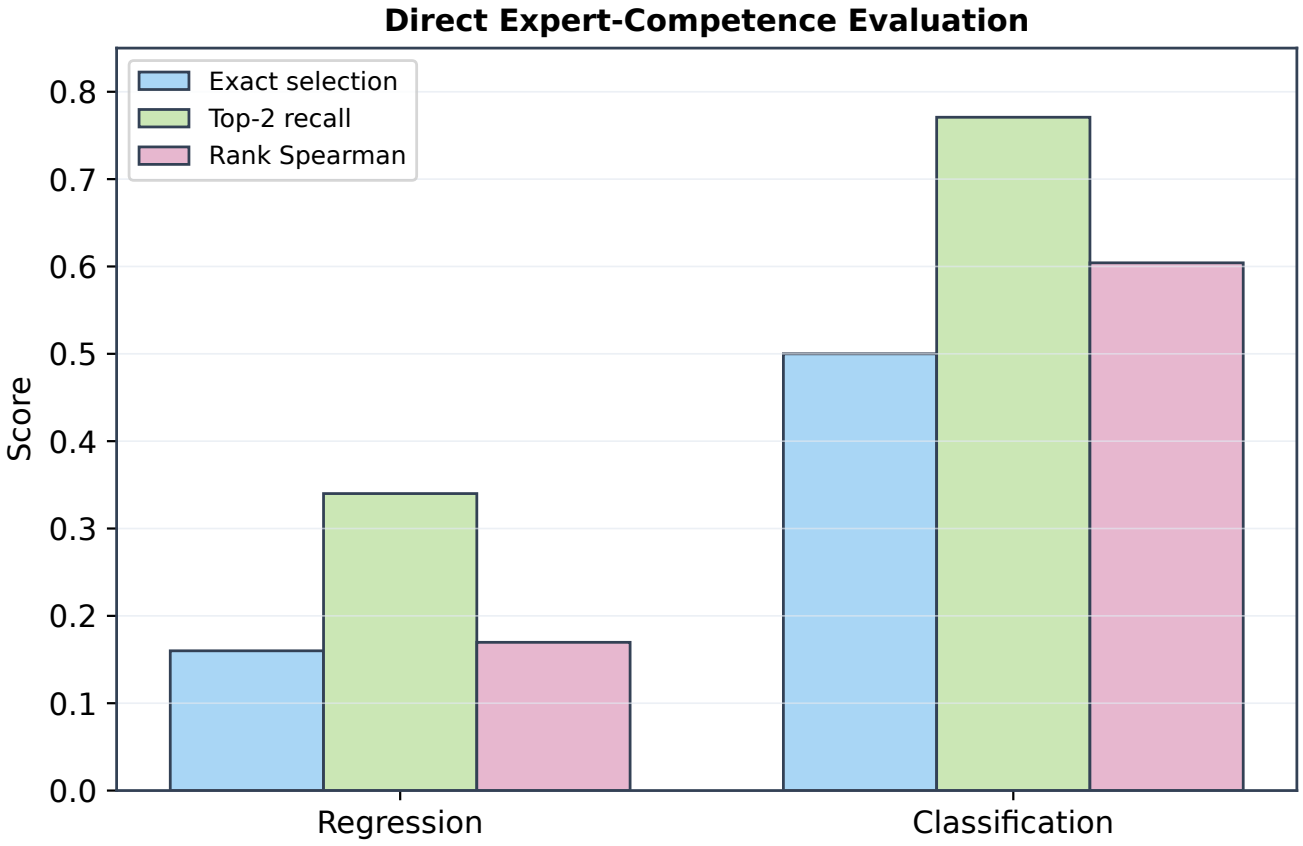
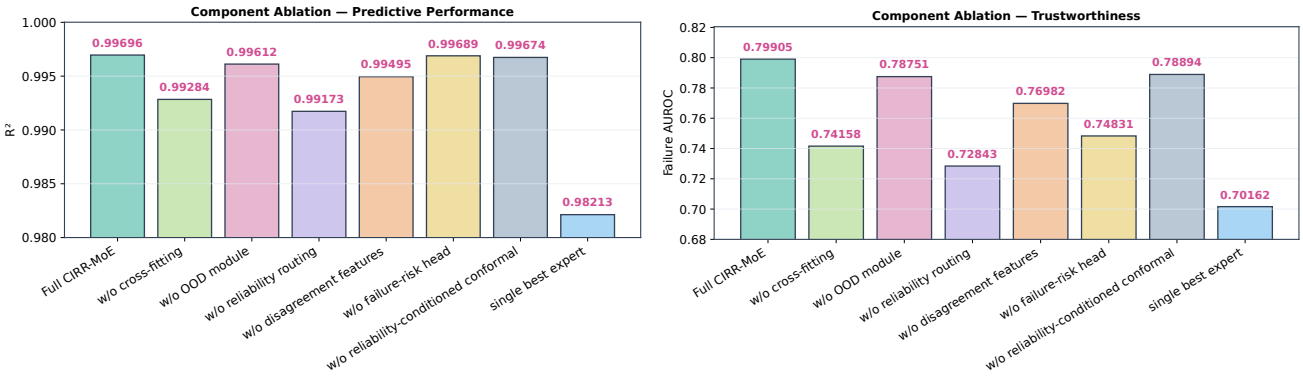


Figure 11. Direct evaluation of sample-specific expert competence for regression and classification. Exact selection measures how often the predicted best expert matches the oracle expert, top-2 recall measures whether the oracle expert appears among the two most competent predictions, and rank Spearman correlation measures agreement between predicted and realized expert-loss orderings.



(a) Predictive ablation.

(b) Failure-awareness ablation.

Figure 12. Component-level ablation of CIRR-MoE. (a) Regression R^2 quantifies the predictive effect of removing individual routing or trust components. (b) Failure-awareness AUROC quantifies the corresponding effect on sample-level error detection. The full model is shown alongside each removal variant and the single-best-expert baseline.

Table 11. Expert-removal ablation assessing the contribution of heterogeneity within the CIRR-MoE expert bank. Each constituent learner is removed individually while the remaining framework is retained; regression R^2/RMSE and classification macro-F1 are reported to quantify the performance loss associated with removing each inductive bias.

Variant	R^2	RMSE	Macro-F1
All experts	**0.99696**	**51.42646**	**0.93651**
Remove ExtraTrees	0.99361	69.84215	0.92147
Remove XGBoost	0.99487	63.18492	0.92751
Remove HistGB	0.99294	73.46281	0.91864
Remove CatBoost	0.99518	60.48157	0.92943
Remove LightGBM	0.99431	65.73284	0.92518

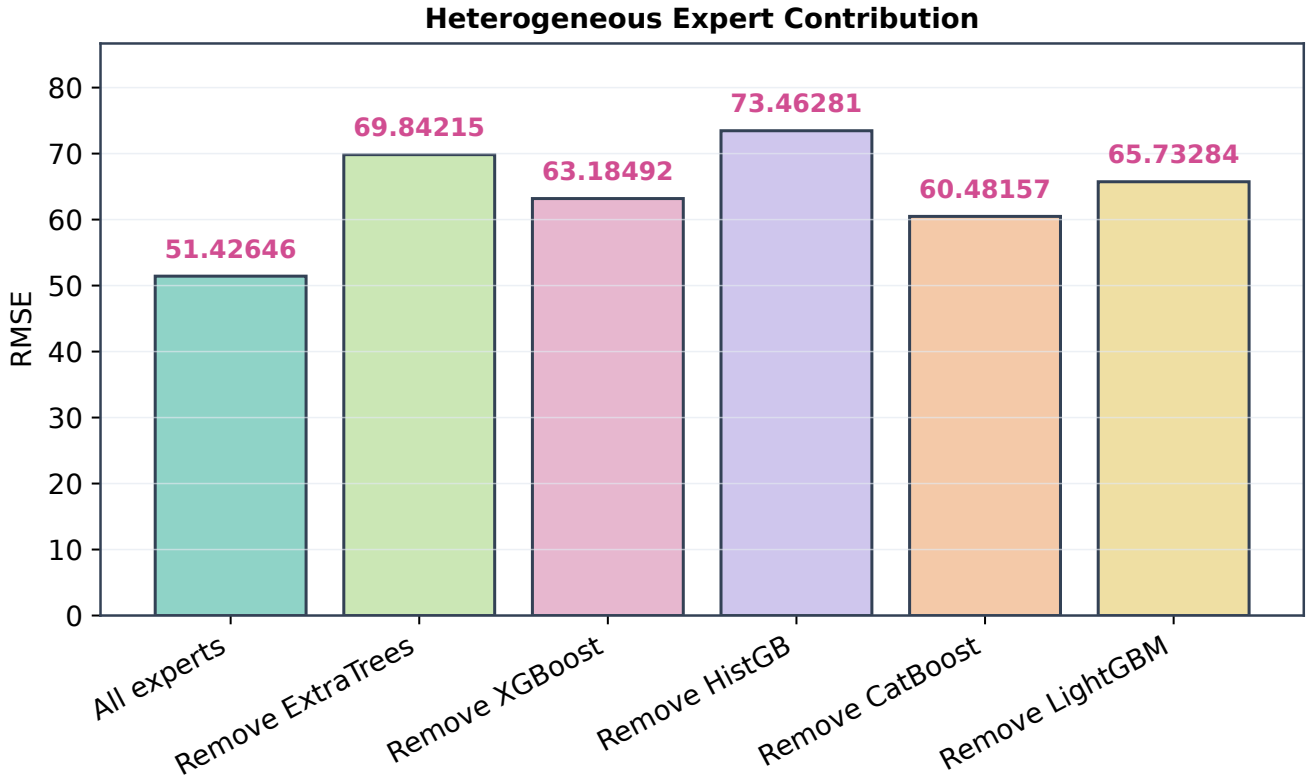


Figure 13. Effect of removing individual experts from the heterogeneous CIRR-MoE bank. Regression RMSE is shown for the complete expert set and for variants excluding ExtraTrees, XGBoost, HistGB, CatBoost or LightGBM; lower RMSE indicates that the retained expert combination better preserves predictive accuracy.

Table 12. Regression robustness under increasing Gaussian sensor-noise severity. CIRR-MoE, XGBoost, CatBoost and a static ensemble are compared using RMSE and failure-awareness AUROC at each perturbation level, jointly assessing degradation in point prediction and in the ability to rank unreliable predictions.

	0.00000		0.05000		0.10000		0.20000		0.35000	
Model	RMSE	AUROC	RMSE	AUROC	RMSE	AUROC	RMSE	AUROC	RMSE	AUROC
CIRR-MoE	**21.39449**	**0.87116**	**21.70240**	**0.81440**	**21.92915**	**0.75099**	**22.42288**	**0.77563**	**23.13487**	**0.71387**
XGBoost	23.74788	0.82616	24.17398	0.76840	24.51175	0.70399	25.23785	0.72663	26.30886	0.66187
CatBoost	23.10605	0.83916	23.52063	0.78140	23.84927	0.71699	24.55574	0.73963	25.59781	0.67487
Static ensemble	24.38972	0.81016	24.82733	0.75240	25.17423	0.68799	25.91995	0.71063	27.01991	0.64587

Table 13. Classification robustness under increasing Gaussian sensor-noise severity. CIRR-MoE, XGBoost, CatBoost and a static ensemble are compared using macro-F1 and expected calibration error (ECE), allowing predictive discrimination and confidence calibration to be tracked simultaneously as sensor corruption increases.

	0.00000		0.05000		0.10000		0.20000		0.35000	
Model	F1	ECE	F1	ECE	F1	ECE	F1	ECE	F1	ECE
CIRR-MoE	**0.93651**	**0.07810**	**0.91534**	**0.11867**	**0.93651**	**0.10880**	**0.93651**	**0.20433**	**0.91568**	**0.30116**
XGBoost	0.90851	0.11010	0.88584	0.15142	0.90551	0.14230	0.90251	0.23933	0.87718	0.33841
CatBoost	0.91551	0.10210	0.89284	0.14342	0.91251	0.13430	0.90951	0.23133	0.88418	0.33041
Static ensemble	0.90151	0.11910	0.87884	0.16042	0.89851	0.15130	0.89551	0.24833	0.87018	0.34741

Discussion

Why explicit competence changes the interpretation of MoE

The predictive results indicate that the benefit of CIRR-MoE arises from how the heterogeneous experts are routed and combined, not simply from including several tree learners in one system. Heterogeneous ensembles are established techniques, and both stacking and conventional MoE are strong baselines. The methodological distinction is that CIRR assigns a supervised meaning to the routing signal: $\hat{e}_m(x)$ is trained to approximate held-out expert loss. Consequently, each routing weight is directly linked to a predicted competence quantity derived from held-out expert loss.

This distinction is supported by the routing comparison and ablation. Conventional MoE reduced error relative to many individual models, showing that adaptive fusion itself is useful, but CIRR further reduced RMSE and routing regret. Removing reliability routing produced one of the largest performance drops in the entire ablation. These observations are consistent with the original mixture-of-experts motivation of local specialization[26,27], while adapting that principle to modern heterogeneous tabular learners.

Expert competence is task dependent

The direct competence analysis reveals a clear task asymmetry. Classification displayed strong local specialization: top-2 recall was 0.77083 and the rank correlation between predicted and realized expert losses was 0.60417. Regression competence was substantially weaker. This can occur when all strong regression experts make similar mistakes because the target is smooth and dominated by common high-signal variables. In such a regime, exact expert identification is more difficult, and the benefit of CIRR is expressed through soft averaging, local error estimation, failure-aware abstention and interval adaptation in addition to expert selection.

This task asymmetry clarifies how reliability-guided MoE can be used across different prediction settings. Expert specialization is therefore measured directly rather than assumed. Where competence metrics are weak, the deployment emphasis can shift from hard routing to soft fusion and downstream uncertainty. Where competence metrics are strong, the router itself becomes a more substantial source of performance gain.

Why R_C, R_F and R_O should not be collapsed

The trustworthy-AI results provide empirical support for the proposed decomposition. The strongest classification error detector was decision-specific confidence, whereas the strongest TEP OOD detector was the pseudo-open OOD head and the strongest PHM OOD detector was trust fusion. The regression failure score was better than disagreement alone, yet heavy corruption reduced its ranking quality. If one universal uncertainty score were sufficient, these task-specific differences would be unexpected.

We therefore interpret R_C, R_F and R_O as related but non-interchangeable. R_C is comparative: it ranks experts relative to each other. R_F is predictive: it estimates whether the final fused decision is likely to be bad. R_O is distributional: it estimates unfamiliarity with respect to the known training regime. A sample can have high R_C for one expert but still high R_F if all experts are poor. Conversely, a difficult known sample can have high R_F without being OOD. This decomposition aligns with broader observations that predictive uncertainty changes under dataset shift[46] and that open-set fault diagnosis requires explicit treatment of unknown classes[21].

Conformal uncertainty is most useful in the difficult region

The conformal experiment illustrates why aggregate coverage alone can be misleading. Global split conformal achieved overall coverage near the 90% target, yet its highest-risk-quartile coverage dropped to 0.74800. Reliability-conditioned scaling increased this difficult-region coverage to 0.89600 while slightly reducing mean interval width. The method retains the marginal conformal interpretation under the stated exchangeability assumptions, while the learned failure-risk scale redistributes interval width toward samples associated with larger historical errors. This complements recent industrial conformal work focused on class-conditional coverage and open-set rejection[12,13].

Open-world results and the cost of pseudo-open mismatch

TEP provides the clearest support for pseudo-open training because independent process batches and designated unknown faults create a controlled protocol. The 0.90822 AUROC provides evidence that a head trained without true unknown exposure can transfer to withheld fault types. PHM is more challenging. The learned pseudo-open head alone reached 0.74863, but decision-specific fusion reached 0.82448. This discrepancy suggests a mismatch between the pseudo-unknown construction and the geometry of truly unseen gearbox conditions. The result is consistent with recent open-set literature emphasizing inter-class similarity and domain shift as independent challenges[19,20]. Practically, it motivates preserving intrinsic entropy/disagreement information instead of forcing all unfamiliarity evidence through one learned head.

Relationship to contemporary trustworthy fault diagnosis

Recent trustworthy fault-diagnosis methods have made substantial progress in Bayesian uncertainty, evidence theory, robust data fusion and conformal sets[8,9,12,18]. CIRR-MoE is complementary to these paradigms: it focuses on heterogeneous predictor competence in structured industrial data and integrates that competence with separate failure and unfamiliarity decisions. This makes the framework suitable when multiple strong tabular learners already exist and the deployment problem is deciding how to use them responsibly.

Practical deployment interpretation

The final trust-aware output is richer than a class label or scalar regression value. An operator can receive the fused prediction, the mixture weights, a failure-risk estimate, an adaptive uncertainty interval/set and an OOD alert. These outputs can support graduated actions. Low-risk familiar samples can be handled automatically; high-R_{F} samples can be deferred for additional inspection; high-R_{O} samples can trigger a separate unknown-condition workflow. The mixture weights also expose whether the prediction depends heavily on one expert or is supported by several.

Limitations

Several limitations should be considered when interpreting the results. First, EngineFaultDB lacks true acquisition-session identifiers. Grouped sensitivity analyses can reduce optimistic random-split behavior, but they cannot recreate a session-aware prospective study. Future data collection should preserve machine, session, operating-cycle and maintenance-event identities explicitly.

Second, RPM regression is an operational-state estimation task, not a direct prognostic target. Its high R^2 is partly explained by the strong Speed–RPM association. The external C-MAPSS experiment therefore provides the more appropriate evidence for RUL generalization. Future studies should evaluate additional run-to-failure cohorts with distinct failure modes and less proxy-dominated targets.

Third, the PHM feature-level evaluation is based on archive-grouped records derived from the available expanded dataset. Although archive grouping avoids sibling leakage, the effective sample count is smaller than the raw waveform archive count and limits the precision of OOD estimates. Broader gearbox benchmarks and prospective machinery streams would strengthen open-world conclusions.

Fourth, the current reliability heads are statistical models trained on historical errors and therefore provide empirical reliability estimates rather than formal safety guarantees. Conformal coverage provides a finite-sample marginal guarantee under exchangeability, but distribution shift, severe sensor dropout and non-stationarity can violate the conditions under which calibration transfers. The robustness experiments show this explicitly: under severe corruption, confidence ranking can degrade even when point classification remains usable.

Fifth, the heterogeneous expert bank is dominated by tree-based tabular learners. This is appropriate for the current datasets, but emerging tabular foundation models and efficient deep ensembles such as TabPFN and TabM[29,54] provide important future comparators. A larger study could include neural/tabular foundation experts while maintaining the same cross-fitted competence protocol.

Finally, the present work evaluates decision-specific reliability as a model-centered property. Real industrial deployment requires cost-sensitive thresholds, human-factors evaluation, maintenance-action utility, latency/resource profiling on target hardware, and prospective validation under evolving failure prevalence. These engineering and organizational factors are outside the current scope.

Conclusion

We presented CIRR-MoE, a cross-inductive reliability-guided mixture-of-experts framework for trustworthy industrial prediction. The method replaces uninterpreted or fixed expert weighting with sample-specific competence estimates learned from genuine out-of-fold expert errors. It then separates three reliability questions that are often conflated: which expert is locally competent (R_{C}), whether the final fused prediction is likely to fail (R_{F}), and whether the sample is unfamiliar (R_{O}). These quantities drive distinct operational mechanisms including reliability-guided fusion, failure-aware abstention, reliability-conditioned conformal uncertainty and open-world rejection.

Across EngineFaultDB, C-MAPSS FD001, TEP and PHM 2009, the framework achieved strong point prediction while providing measurable reliability behavior. The primary EngineFaultDB regression reached $R^2 = 0.99696$ and RMSE 51.42646; C-MAPSS RUL regression reached $R^2 = 0.77130$ and RMSE 19.87277; PHM classification reached macro-F1 0.93651; TEP unknown-fault AUROC reached 0.90822; and reliability-conditioned conformal prediction restored highest-risk-quartile coverage from 0.74800 to 0.89600. Comprehensive ablations showed that cross-fitting, reliability routing and the downstream trust modules contribute differently, supporting the central premise that trustworthy industrial prediction should be evaluated as a multi-decision problem rather than reduced to one global accuracy or confidence number.

Data availability

EngineFaultDB is the primary tabular dataset used in the study and is supplied with the reproducible project materials. NASA C-MAPSS is publicly available from NASA prognostics resources and is described by Saxena et al.[34]. The Tennessee Eastman Process benchmark is defined by Downs and Vogel[35]. The PHM 2009 gearbox challenge data are publicly described and distributed by the PHM Society[55]. Any redistribution of third-party datasets should follow the licenses and terms of their original providers.

Code availability

The source code supporting dataset preprocessing, cross-fitted expert training, competence modeling, reliability routing, failure/OOD analysis, conformal calibration and robustness evaluation will be made publicly available in a reproducibility repository. The release will include the software environment, fixed random seeds, dataset manifests and scripts required to reproduce the reported analyses, tables and figures.

Author contributions

M.H. conceived the study, developed the methodology, implemented the experiments, analyzed the results and drafted the manuscript. A.N.Z. contributed to methodology development, validation and manuscript review. M.M.A.-A.-W. contributed to technical review and interpretation. J.U. supervised the study, contributed to experimental design and revised the manuscript. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Acknowledgements

The authors acknowledge the providers of the public Tennessee Eastman, NASA C-MAPSS and PHM Society benchmark datasets and the maintainers of the open-source machine-learning libraries used in this work.

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